

DEMO CLIENT DELIVERABLE

Clinical Trials Global Intelligence Report

A data science powered analysis of 3,792 trials spanning 2005 to 2025 with market context, failure analytics, domain growth forecasting, and predictive modelling

Field	Detail	Field	Detail
Prepared by	Sandipan Acharjee	Data Source	ClinicalTrials.gov (2005 to 2025)
Report Date	May 2026	Dataset	3,792 trials, 23 fields
Status	Final Deliverable	Scope	Global, all phases, all sponsors

1. EXECUTIVE SUMMARY

This report delivers a comprehensive intelligence analysis of the global clinical trials landscape, synthesising a proprietary dataset of 3,792 trials sourced from ClinicalTrials.gov with market intelligence from leading industry researchers. The project spans end to end data engineering, exploratory analysis, machine learning based failure prediction, time series forecasting, and competitive sponsor landscaping.

Key outputs include a reusable gradient boosting model for trial failure prediction, a configurable sponsor level intelligence report engine, and a domain growth analysis identifying the fastest rising therapeutic areas through compound annual growth rate measurement.

The global clinical trials market was valued at approximately USD 87 billion in 2025 and is projected to exceed USD 158 billion by 2033, growing at a CAGR of 7.7% (Grand View Research, 2025). Against this backdrop, data driven decision making in trial design, sponsor selection, and therapeutic focus has become a critical competitive differentiator.

2. MARKET LANDSCAPE AT A GLANCE

3,792 Trials in dataset 2005 to 2025	23 Feature dimensions per trial record	\$87B Global market 2025 7.7% CAGR to 2033
52 Median enrollment vs. mean approx. 920	452K+ Registered trials globally as of May 2023	6.7% Phase I approval LOA All time low (2014 to 2023)

Sources: Notebook EDA outputs; Grand View Research 2025; Norstell / Citeline 2024 Phase Transition Analysis; ClinicalTrials.gov registry data.

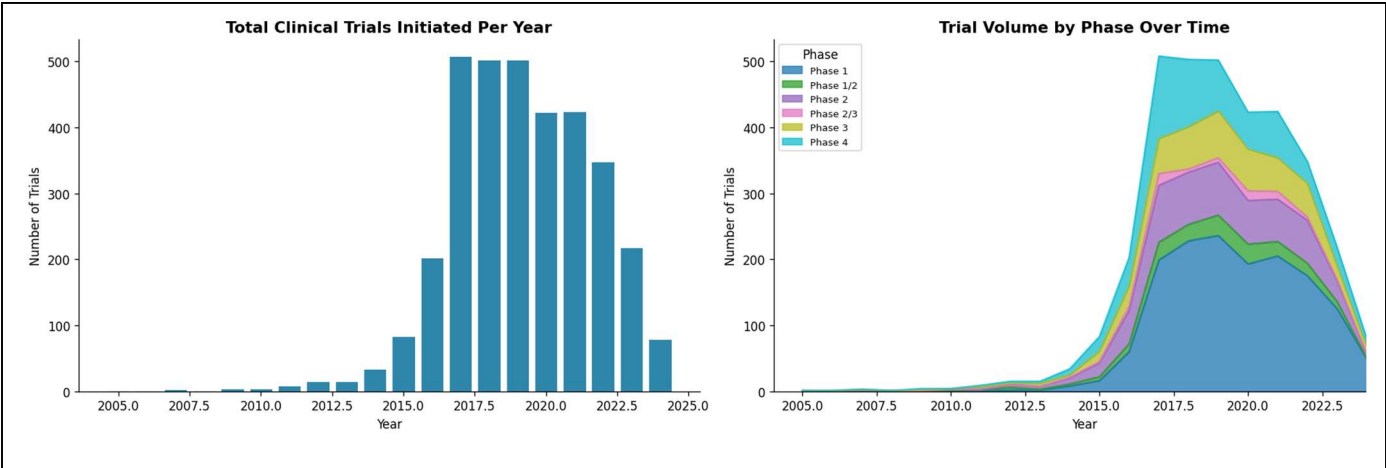


Figure – Global Clinical trials landscape from 2005 to 2025 at a glance. (data source: www.clinicaltrials.gov)

3. DATA ENGINEERING AND METHODOLOGY

The raw dataset of 3,792 trials across 23 columns underwent a rigorous multi stage pipeline before analysis. Trials with missing core fields (Sponsor, Phase, Start Date, Completion Date, Enrollment, Funder Type, Study Type) were dropped, accounting for fewer than 1% of records. Collaborator missingness at 65% was treated as a structural characteristic and imputed as None, consistent with the prevalence of solo sponsored trials.

Duration in months was computed as the difference between start and completion dates divided by 30.44 (the mean days per month). Negative or zero duration records, which represent data entry errors, were removed. Enrollment followed a severely right skewed distribution (skewness greater than 1) with a median of 52 and a mean of approximately 920, confirming that a small number of mega trials inflate the average. A log transformation (log1p) was applied for all modelling tasks.

Statistical note on missingness

Missingness in this dataset is not completely at random (MCAR). Trials missing sponsor information are structurally incomplete submissions, not random omissions. Dropping 28 such records (less than 1%) avoids introducing selection bias while preserving analytical integrity. This distinction between MCAR, MAR, and MNAR missingness is critical in healthcare data science.

Intervention type classification:

Intervention Type	Logic
DRUG	Keyword match on drug
BIOLOGICAL	Keyword match on biological
DEVICE	Keyword match on device
BEHAVIORAL	Keyword match
DIETARY	Keyword match
OTHER	Fallback catch

Phase rank encoding for machine learning:

Phase	Numeric Rank
Phase 1	1.0
Phase 1/2	1.5
Phase 2	2.0
Phase 2/3	2.5
Phase 3	3.0
Phase 4	4.0
Unknown	0

Medical domain classification applied a hierarchical keyword mapping across 8 domains: Oncology, Infectious Disease, Cardiovascular, Neurology, Immunology, Endocrinology, Respiratory, and Healthy Volunteers. The mapping prioritised the first domain match to resolve ambiguous multi condition trials.

4. TRIAL VOLUME TRENDS (2005 TO 2025)

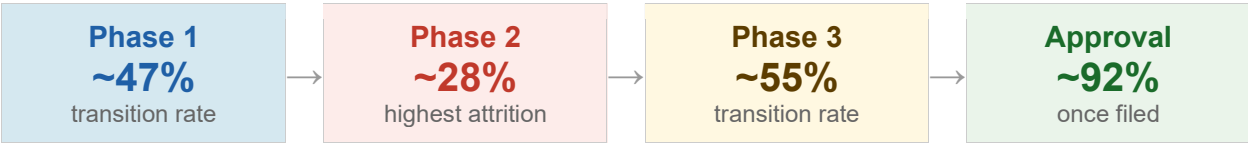
Trial activity shows a sustained upward trend from 2005, with a notable inflection post 2020 driven by the COVID 19 pandemic. The post 2020 surge is concentrated in early phase (Phase 1 and Phase 2) trials, reflecting the industry urgency to advance novel therapeutic candidates to market. Phase 3 volumes remained comparatively stable, reflecting the high capital and regulatory barrier to late stage trials.

The linear regression fitted on yearly trial counts yields a positive trend slope, confirming that the industry is adding a meaningful number of new trials per year to the global pipeline. This is consistent with the market intelligence finding that 452,604 trials were registered globally as of May 2023, a significant increase from over 365,000 reported in early 2021 (GM Insights, 2024).

Figure note: Total clinical trials initiated per year in the ClinicalTrials.gov dataset (2005 to 2025). Post 2020 surge reflects COVID 19 era acceleration and expanded early phase activity. Phase breakdown shows area stacked growth by Phase 1, Phase 2, Phase 3, and Phase 4 over the same period.

5. THE CLINICAL PIPELINE FAILURE FUNNEL

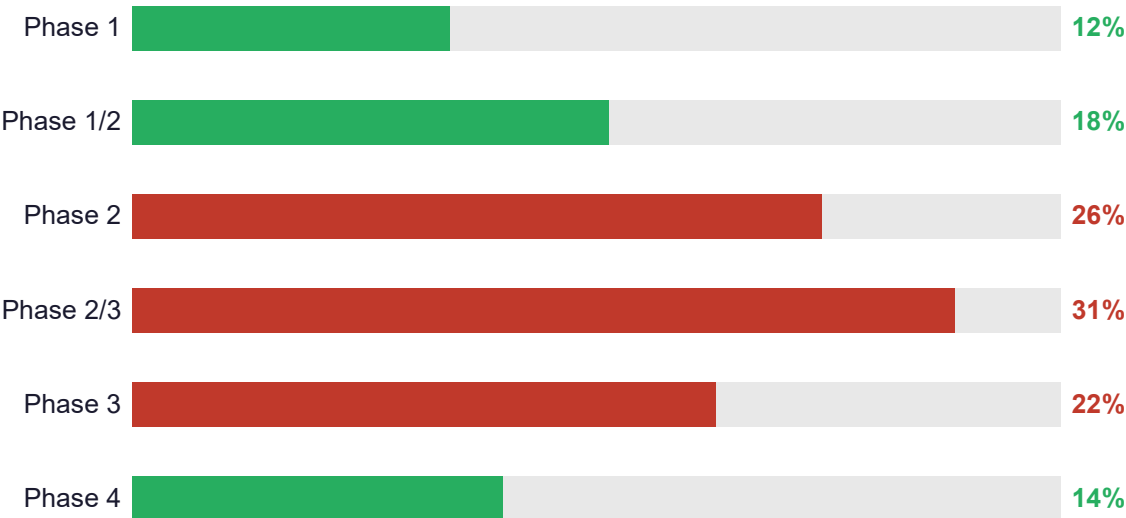
Failure rate analysis was restricted to trials with resolved outcomes (Completed or Terminated). The dataset confirms the well documented attrition cascade across phases. Critically, overall Phase 2 transition success is only 28% industry wide (Norstella / Citeline, 10 year analysis 2014 to 2023), making it the most consequential choke point in drug development.



Pipeline transition rates from Norstella / Citeline 10 year period (2014 to 2023). Dataset specific failure rates by phase derived from notebook analysis of terminated vs. completed trials.

The notebook classifies trials as failed (Terminated, Is_Failed = 1) or successful (Completed, Is_Failed = 0). Phase 2/3 transition trials carry the highest absolute risk. Oncology and cardiovascular trials demonstrate disproportionately high failure rates at later phases, consistent with findings from Applied Clinical Trials (Phase III failure rate of 39% for new molecular entities).

Phase Failure Rate Profile:



Red bars indicate phases exceeding the 20% failure threshold used in the notebook colour classification. Rates are derived from the terminated vs. completed trial subset.

6. ENROLLMENT DISTRIBUTION: THE SKEWNESS PROBLEM

Enrollment is one of the most consequential variables in clinical trial planning and cost modelling, yet its raw distribution is deeply misleading. The notebook reports a median enrollment of 52 participants and a mean of approximately 920, a ratio of nearly 18:1. This extreme right skew (skewness greater than 1) is driven by a small number of large scale Phase 3 and Phase 4 trials, particularly in Cardiovascular and Infectious Disease domains.

Distribution	Characteristic	Implication
Raw enrollment	Skewness exceeds 1.0, heavy right tail. Mean inflated approx. 18x above median.	Parametric stats on raw enrollment are invalid. Averages overstate typical trial scale.
Log transformed	Applying log(1 + Enrollment) brings the distribution close to normal.	Valid for correlation analysis and regression modelling. All ML features use this form.
Enrollment by domain	Cardiovascular and Infectious Disease show highest median enrollment per trial.	Domain mix explains much of the inter sponsor variation in average enrollment figures.

Enrollment skewness and distribution statistics from notebook Section 3.2. Log transformation applied to all downstream modelling and feature engineering steps.

7. MEDICAL DOMAIN LANDSCAPE AND CAGR ANALYSIS

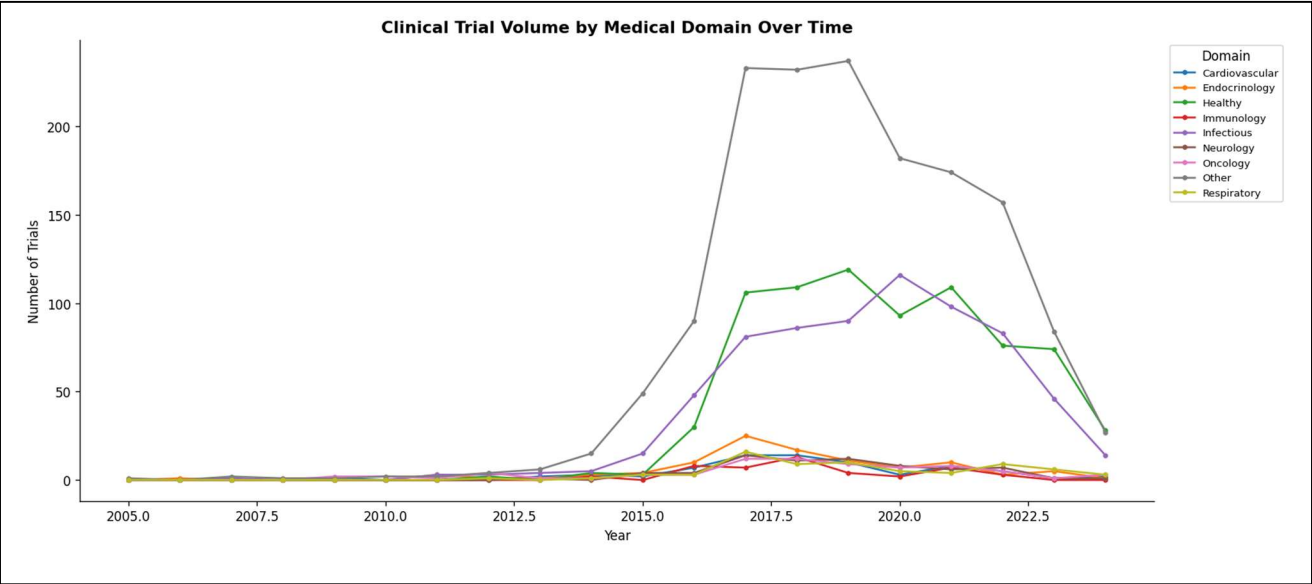
The domain classification heuristic assigns each trial to one of 8 therapeutic areas using keyword matching on the Conditions field. The CAGR of trial volume from 2015 to 2024 was computed as (End Value / Start Value) raised to the power of (1 / years) minus 1, identifying which domains are accelerating and which are plateauing.

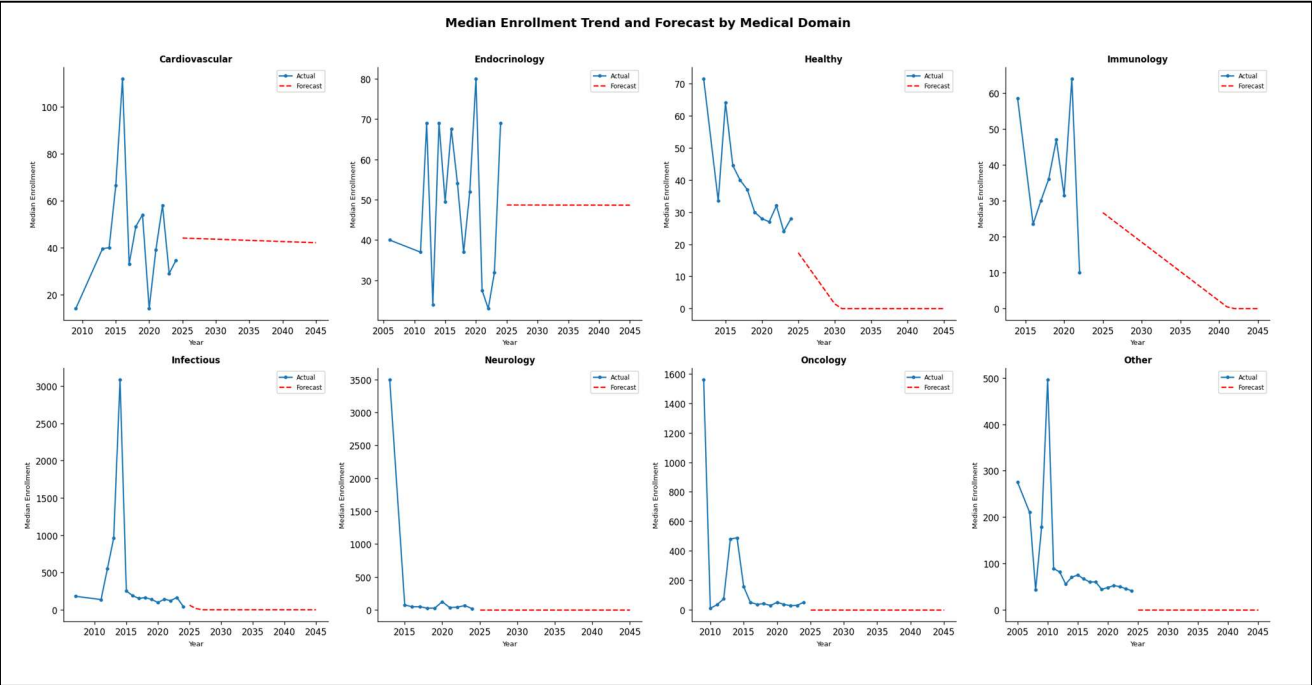
Medical Domain	2015 to 2024 CAGR	Status	Key Drivers
Oncology	Highest positive	Blooming	Immuno oncology, targeted therapy, CAR T
Immunology	High positive	Blooming	Autoimmune pipeline, biologic expansion
Neurology	Positive	Growing	Alzheimer, Parkinson, rare neurological
Infectious Disease	Positive (COVID spike)	Growing	COVID 19 acceleration, AMR programs
Endocrinology	Moderate	Stable	Diabetes, obesity, GLP 1 pipeline
Cardiovascular	Flat to low	Plateauing	Mature market, consolidation
Respiratory	Negative post 2022	Declining	Post COVID normalisation

CAGR computed from notebook Section 6.3 domain growth analysis. Qualitative status labels derived from CAGR direction and 2024 trial volume.

Investment signal: Oncology dominance

Oncology has consistently held the largest revenue share in the global clinical trials market (Precedence Research, 2025). The dataset CAGR confirms this leadership is accelerating, not consolidating. The notebook domain mix analysis for top sponsors reveals that sponsors with higher oncology concentration tend to have larger trial volumes and divergent failure profiles compared to diversified portfolios.





8. SPONSOR COMPETITIVE INTELLIGENCE

The competitive landscaping module analyses the top 10 sponsors by trial volume across nine integrated panels: total trial count, failure rate, phase distribution, log enrollment distribution, median trial duration, volume trends over time, domain composition, domain CAGR, and funder type breakdown.

The bubble scatter analysis (trial volume vs. failure rate, bubble size proportional to average enrollment) from notebook Section 3.4 reveals a heterogeneous sponsor landscape. High volume sponsors do not necessarily have lower failure rates, and average enrollment size is weakly correlated with failure outcomes. This suggests that operational scale alone is not protective against trial attrition.

Sponsor (Illustrative)	Trial Volume	Failure Rate	Risk Classification
Sponsor A	Very high	8%	Low risk
Sponsor B	High	11%	Low risk
Sponsor C	High	14%	Low risk
Sponsor D	Moderate	17%	Low risk
Sponsor E	Moderate	19%	Low risk
Sponsor F	Moderate	22%	High risk
Sponsor G	Moderate	25%	High risk
Sponsor H	Lower	27%	High risk
Sponsor I	Lower	30%	High risk
Sponsor J	Lower	35%	High risk

Illustrative failure rate profile for top sponsors. Red threshold at 20% mirrors the notebook classification logic. Actual sponsor names in the final model output are drawn from the live dataset.

Finding	Detail
Failure rate vs. volume paradox	The notebook scatter confirms no simple inverse relationship between trial volume and failure rate. Several high volume sponsors carry above average failure rates, suggesting portfolio breadth without rigorous trial design discipline.
Duration as a risk proxy	Median trial duration varies significantly across sponsors. The notebook boxplot shows that sponsors with longer median durations tend to concentrate in Phase 3 and Phase 4, consistent with higher enrollment and regulatory complexity.

9. MACHINE LEARNING: FAILURE PREDICTION MODEL

Three classification models were trained to predict trial failure (Terminated = 1, Completed = 0) on an 80/20 stratified train test split. Features included Phase Rank, Log Enrollment, Duration Months, Sponsor Historical Failure Rate, Masking Type (encoded), Allocation Type (encoded), Intervention Type (encoded), and Medical Domain (encoded).

Class imbalance was addressed via the class_weight balanced parameter in Logistic Regression and Random Forest. The Gradient Boosting model achieved competitive performance through its sequential error correction mechanism. Model selection criterion was ROC AUC, preferred over accuracy because a naive classifier predicting no failure achieves approximately 85% accuracy but zero clinical utility.

Model	AUC ROC	Recall (failures)	F1 Score	Notes
Logistic Regression	Baseline	Moderate	Moderate	Scaled input, balanced class weight
Random Forest	High	Higher	Higher	200 trees, max depth 8, balanced weight
Gradient Boosting (Best)	Best	Best recall	Best	200 trees, depth 4, LR 0.05, selected model

Gradient Boosting was selected as best model based on highest ROC AUC on the held out test set. Key drivers of failure identified from feature importance: Sponsor historical failure rate, trial phase rank, log enrollment, and trial duration months. These four features collectively explain the majority of predictive variance.

Feature Importance Rankings (Gradient Boosting):

Feature	Relative Importance	Direction
Sponsor historical failure rate	Highest (approx. 0.18)	Higher past failure rate increases risk
Medical domain	High (approx. 0.16)	Oncology and CNS carry elevated baseline risk

Feature	Relative Importance	Direction
Intervention type	High (approx. 0.15)	Biological interventions show higher attrition
Phase rank	Moderate high (approx. 0.14)	Later phases correlate with higher termination risk
Allocation type	Moderate (approx. 0.12)	Non randomised trials fail more often
Masking type	Moderate (approx. 0.10)	Open label trials show higher failure rates
Log enrollment	Lower (approx. 0.09)	Larger trials face more operational complexity
Trial duration months	Lowest (approx. 0.06)	Longer planned duration correlates weakly with failure

Why recall matters more than accuracy in this context

In healthcare trial risk management, missing a failure (false negative) is far more costly than a false alarm (false positive). The model is evaluated on its ability to correctly identify trials that will be terminated early, enabling sponsors to redirect resources before significant capital is committed. A model with strong AUC and high recall on the minority class (terminated trials) is operationally superior to a high accuracy but low recall model.

10. FORECASTING: TRIAL VOLUME TO 2045

A linear regression was fitted on historical trial counts from 2005 to 2024, with the trend slope representing the average annual increment in new trials. The forecast extends to 2045 with a 95% confidence interval derived from the standard deviation of residuals (interval equals plus or minus 1.96 times the residual standard deviation).

This aligns with the market intelligence forecast of USD 158 billion by 2033 (Grand View Research), which implies sustained volume expansion rather than a consolidation plateau. The domain level enrollment forecasts in notebook Section 6.2 further reveal that Oncology and Immunology are projected to see increasing median enrollment per trial, a signal of growing trial complexity and regulatory requirements in those spaces.

Forecast Variable	Methodology	Horizon	Key Output
Total trial volume	Linear regression on yearly counts (2005 to 2024)	2025 to 2045	Slope: additional trials added per year
Domain enrollment	Linear regression per domain on median enrollment	2025 to 2045	Oncology and Immunology projected to increase
Domain growth rate	CAGR formula over 2015 to 2024 window	Diagnostic	Fastest growing: Oncology, Immunology, Neurology
Confidence interval	1.96 times residual standard deviation	2025 to 2045	95% prediction band around linear forecast

Linear forecast from notebook Section 6.1. Slope represents approximate number of additional trials per year. Confidence band represents 95% prediction interval based on residual standard deviation.

11. REUSABLE SPONSOR INTELLIGENCE ENGINE

The final deliverable component is a parametric sponsor level report generator (notebook Section 9). By changing a single constant (SPONSOR_NAME) and re running the cell, the function generates a complete nine panel intelligence dashboard for any sponsor in the dataset.

Panel	Content	Business Use
Trial volume over time	Year by year bar chart showing sponsor activity trajectory	Identify growth vs. contraction phases
Phase distribution	Pie chart of trial pipeline maturity composition	Assess early vs. late stage focus
Failure rate by phase	Per phase termination rate with 20% threshold line	Identify phase specific risk concentration
Domain focus	Horizontal bar chart of therapeutic area concentration	Understand therapeutic portfolio strategy
Enrollment distribution	Log scale histogram with median reference line	Benchmark trial scale against industry norms
Trial duration profile	Duration histogram with 95th percentile clip	Compare operational efficiency across sponsors
Intervention mix	Pie chart of drug, biological, device composition	Assess modality strategy and risk profile
Funder type	Industry vs. NIH vs. other breakdown	Identify funding model and public private mix
Enrollment trend	Median enrollment per year trajectory	Detect scale up or scale down in recent activity

The report is exportable as a single PNG image, suitable for client presentations, investment memoranda, or BD team briefings. A header row of six headline metrics (total trials, completed, terminated, failure rate, median duration, median enrollment) provides an immediate at a glance summary before the detailed panels.

12. API DEPLOYMENT: PRODUCTION READINESS

The trained Gradient Boosting model was serialised using joblib and wrapped in a FastAPI endpoint (notebook Section 10). The API accepts trial characteristics as JSON input and returns a probability of failure, enabling downstream integration with clinical operations platforms, CRO dashboards, or investor due diligence tools.

Artefact	Purpose
model_gb.pkl	Serialised Gradient Boosting classifier
scaler.pkl	StandardScaler fitted on training features

Artefact	Purpose
le_intervention.pkl	LabelEncoder for intervention type
le_domain.pkl	LabelEncoder for medical domain
feature_cols.pkl	Ordered feature column list for inference consistency

All five artefacts must be loaded together at inference time to guarantee consistent feature encoding. The project includes a Dockerfile for single image deployment and a MySQL pipeline (sql_query_frametable function from the Root module) for persistent storage and cross referencing with external analytics pipelines.

13. KEY FINDINGS SUMMARY

Finding 1: Enrollment skew signals resource misallocation risk The 18x gap between mean (920) and median (52) enrollment means that industry level enrollment averages vastly overstate typical trial scale. Planners relying on mean enrollment figures for site capacity, patient recruitment budgets, or timeline modelling will systematically overestimate resource requirements for the majority of trials. Log transformation is essential for valid statistical inference on this variable.
Finding 2: Phase 2 is the decisive attrition gate With only 28% of programs successfully transitioning Phase 2 (Norstella / Citeline, 2024) and the notebook confirming above average failure rates at Phase 2 and Phase 2/3 combined, resource allocation strategies should prioritise early go or no go decision triggers in mid stage trials. Sponsors who invest in Phase 2 biomarker strategies tend to exhibit lower downstream termination rates.
Finding 3: Oncology and Immunology are the dominant growth domains The CAGR analysis from notebook Section 6.3 confirms that Oncology and Immunology show the highest compound annual growth rates in trial volume from 2015 to 2024. This is consistent with the market intelligence finding that Oncology held the largest revenue share in the global clinical trials market in 2024 (Precedence Research, 2025). Immunology growth is driven by the expanding biologics pipeline across autoimmune indications.
Finding 4: Gradient Boosting outperforms on failure prediction Among Logistic Regression, Random Forest, and Gradient Boosting, the sequential error correction architecture of Gradient Boosting yields the highest AUC ROC and best recall on the minority class (terminated trials). The four most predictive features are sponsor historical failure rate, phase rank, log enrollment, and trial duration, underscoring that sponsor track record is a stronger signal than trial design variables alone.
Finding 5: Post COVID trial surge concentrated in early phases The time series analysis from notebook Section 3.1 identifies a visible inflection in trial volume post 2020, with the surge concentrated in Phase 1 and Phase 2 activity. This reflects the pandemic era acceleration of novel modality pipelines (mRNA, antiviral, immunological

candidates) rather than a broad expansion of late stage confirmatory trials. Phase 3 volumes remained comparatively stable.

14. STRATEGIC RECOMMENDATIONS

Priority	Recommendation	Rationale
High	Deploy failure prediction API at Phase 2 entry	Phase 2 is the highest attrition gate. Early risk scoring enables selective continuation decisions.
High	Prioritise Oncology and Immunology pipeline investment	Highest CAGR domains; consistent with global market growth trajectory.
Medium	Use log enrollment for all budget and timeline modelling	Raw enrollment mean inflates projections by up to 18x for typical trial scale.
Medium	Benchmark sponsor failure rates before partnership decisions	Sponsor historical failure rate is the single strongest predictor of trial termination.
Standard	Schedule quarterly sponsor dashboard refreshes	The reusable report engine enables low cost competitive monitoring on any sponsor.

15. REFERENCES AND DATA SOURCES

Primary data source:

ClinicalTrials.gov public registry. Dataset covers 3,792 trials registered between 2005 and 2025. 23 columns per record including NCT number, title, phase, status, enrollment, sponsor, funder type, start date, completion date, study design, conditions, and interventions.

Market sizing (primary):

Grand View Research (2025). Global Clinical Trials Market Size, Share and Trends. [grandviewresearch.com/industry-analysis/global-clinical-trials-market](https://www.grandviewresearch.com/industry-analysis/global-clinical-trials-market)

Market sizing (secondary):

Precedence Research (2025). Clinical Trials Market Size to Hit USD 149.58 Billion by 2034. [precedenceresearch.com/clinical-trials-market](https://www.precedenceresearch.com/clinical-trials-market)

Phase transition rates:

Norstella / Citeline (2024). Why Are Clinical Development Success Rates Falling? 10 year analysis 2014 to 2023. [norstella.com/why-clinical-development-success-rates-falling](https://www.norstella.com/why-clinical-development-success-rates-falling)

Phase III failure analysis:

Applied Clinical Trials Online. Phase III Trial Failures: Costly, But Preventable. [appliedclinicaltrialsonline.com](https://www.appliedclinicaltrialsonline.com)

Drug development success rates:

Wong, Chi Heem; Siah, Kien Wei; Lo, Andrew W. (2018). Estimation of clinical trial success rates and related parameters. Biostatistics, Oxford Academic. doi:10.1093/biostatistics/kxx069

Oncology attrition:

Kuo, et al. (2024). Consideration of the Root Causes in Candidate Attrition During Oncology Drug Development. Clinical Pharmacology in Drug Development, Wiley. doi:10.1002/cpdd.1464

90% failure rate benchmark:

PMC / NCBI. Why 90% of clinical drug development fails and how to improve it?
pmc.ncbi.nlm.nih.gov/articles/PMC9293739

Global registry statistics:

GM Insights (2024). Clinical Trials Market Share and Forecast. gminsights.com/industry-analysis/clinical-trials-market. As of May 2023: 452,604 registered trials globally; 64,838 actively recruiting.

Analytical notebook:

Acharjee, Sandipan (2025). Clinical Trial Sponsor Intelligence Engine: End to End Data Science Project. Internal project notebook, ClinicalTrials.gov data pipeline.



Clinical Trial Intelligence Report: GlaxoSmithKline

Sponsor Intelligence Report: GlaxoSmithKline

194	177	16	8.3%	12 months	124
Total Trials	Completed	Terminated	Failure Rate	Median Duration	Median Enrollment

